

DATA 505 - Statistics Using R

Fall 2025

Instructor information: Andrew Bean, Ph.D. (Adjunct Professor, Department of Mathematics and Computer Science)

Lectures: Tuesday 6:00 - 8:45 pm, Brother's College 21

Office Hours: virtual and by appointment only via Google calendar

<https://calendar.app.google/z4xbpyyQzaEk5Fz66>

Course Description. Covers how to use real-world data and draw intelligent conclusions on it using univariate and bivariate descriptive statistics such as centrality, spread, and correlation. Students have the opportunity to learn how to use the R language in the R-studio environment to obtain representative samples from large datasets and use the information in such samples to draw inferences on the larger datasets they were taken from. Course topics include graphical and tabular presentation of data, descriptive summary statistics, manipulation of data, regression, probability distributions, t-tests, and one-way analysis of variance.

Course Prerequisites. Students are expected to have taken courses in programming and in statistics.

Learning Outcomes. By the end of the course, students will be able to:

1. Select, create, and interpret appropriate graphical displays, descriptive statistics, and inferential statistics. They will apply the concepts of robustness and model validation;
2. Analyze real data within authentic applications;
3. Assess the quality of a statistical argument;
4. Use the R programming language;
5. Interpret results in language that is both statistically correct and understandable by someone who has not taken a course in statistics;
6. Apply statistical reasoning in broad areas of application including the social, biological, and physical sciences.

Text. R for Data Science (2nd edition) by Garrett Grolemund and Hadley Wickham, <https://r4ds.hadley.nz/>. This book is available online for free. You can also purchase a hard copy if you prefer. Additional readings will be assigned as needed.

Moodle. The course Moodle site will be used to post all course material and grades.

Laptop Computer and Statistical Software. We will be using the software package R and the R- studio environment. These are open-source and as such may be downloaded and installed for free. Please visit <https://posit.co/download/rstudio-desktop/> first to install R, then to install RStudio. Please see the startup document on Moodle for full instructions.

The installation of the software needs to be completed by the end of class Tuesday, August 26. Try to install the software as soon as you can, so you have time to get help, if necessary. If you have any trouble, please contact the University Technology Service Center Helpdesk. Contact information and hours may be found here: <https://drew.edu/about/technology-resources/university-technology/contact-us/>.

Please log into R and R-studio every day before class; you will be asked to access it and perform small coding assignments during class. Try to use the same computer for homework that you will use for quizzes and exams and regularly practice a variety of problems using R; if you do this, you will be much less likely to experience computer problems during the tests. Should you have a problem with your computer and it was purchased through Drew, take it immediately to the Helpdesk. When you hand in your laptop for repairs, you may request a free loaner. If you purchased your computer elsewhere, you need to arrange for the repair, but

you may rent a loaner while your computer is being repaired. Details of the loaner policy may be found at <https://www.drew.edu/university-technology/policies/loaner-policy/>.

Electronic devices. You are expected to bring a working laptop to each lecture. You are allowed to use an iPad (or similar devices) for class-related activities (such as taking notes). You are **prohibited** to use cell phones, headphones, and earbuds during all lectures. You are also prohibited to use electronic devices for unrelated activities, such as browsing internet, messaging, gaming, working on assignments from a different course, etc.

Attendance. Class attendance and participation are integral to the academic experience at Drew University. As such, you are expected to attend every class unless you have a legitimate excuse, such as illness, family emergency or participation in a college sponsored event. Attendance is mandatory for successful completion of the course, whether you are taking a class in person or online. Students who face challenges participating at the scheduled time should contact the instructor immediately to make alternate arrangements. For legitimate absences, to the extent possible, students should watch recordings, study the PowerPoint slides, and submit due assignments. Students are responsible for class attendance and for the prompt and regular performance of all assigned work. Faculty members are not obligated to review class material, give makeup examinations, or make special arrangements to accommodate absences.

Use of Zoom: This class is not designed to be virtual (e.g., group activities, data-analysis labs). For that reason, I will NOT be recording lectures. Please be aware that attending any class on Zoom will likely impact your learning in a negative way, and you may only do so with a legitimate reason and with prior permission.

Canceled Class. If class is canceled for any reason on an exam date, the exam will be held on the next class period.

Email. Check your email regularly, since assignments and important announcements (such as cancellation of class) are delivered via email. It is every Drew student's responsibility to regularly check his/her email and to not allow the email box to completely fill up (once filled, no new incoming messages can be received). Claiming to have missed an announcement or an assignment because you did not receive the email due to a full mailbox is not an acceptable reason to be excused or extended more time to complete the work.

Office Hours. Please feel free to see me during office hours about any difficulties with the course before it gets too close to an exam date. If the office hours conflict with your course schedule, alternate meetings can be arranged by special request. Note that problems faced early are much easier to solve! Often a new idea learned takes time for its meaning and implications to settle in; allow yourself enough time to digest it properly. NOTE, however, that poor attendance without a valid reason is not an excuse to misuse office hours. Office hours are intended as an opportunity to ask questions about the material and homework after time has been spent on your own with the material in question; they are not make-up lectures or tutoring sessions. Students who routinely miss lectures and expect to make up the time during office hours will be disappointed. Appointments for office hours can be scheduled via Moodle.

Late work and Missed Assignments. There are no make-ups in this class, even for excused absences. Missed or late assignments will earn a zero. If you know in advance that you will miss a class, please let me know as soon as possible. If you know in advance that you will miss a quiz or exam and you let me know ahead of time, you may be able to take it early by making appropriate arrangements with me. Students unable to submit work on time as a result of a documented extended illness should contact their instructor and Dana Giroux in the Office of Accessibility Resources (OAR) to coordinate appropriate accommodations.

Homework. There will be a homework assignment for this course most weeks. The homework

assignments are mandatory; it is imperative that you attempt to do all of them, as they are instrumental to your understanding of the course material. Homework assignments are not graded per se. However, you may be called upon to show your work when homework assignments are discussed in class and your performance at that time will count towards your participation grade.

Components of Your Grades:

15% Attendance and Participation

You are expected to attend class as much as you can, and actively participate during class. Attending class does not simply mean showing up and extremely poor participation during class will also cause your attendance grade to be lowered. Examples of poor participation include arriving late, leaving early, checking your phone during class and using your computer inappropriately. You will also occasionally be called upon to present your homework, your performance on which will be factored into your participation grade.

30% Quizzes and assignments

There will be a quiz or assignment approximately once every 2 or 3 weeks; quizzes will be about 30 minutes during class. Due dates for assignments will be made known when the assignment is set.

25% Midterm

30% Final Exam

Collaboration. No collaboration is permitted on exams and quizzes and the only programs you are allowed to run during exams and quizzes are R and RStudio. You are not allowed to have any cell phone out during quizzes and cell phones and smart watches must be at the front of the room if you bring them to exams. You are not allowed to reference any papers, websites, or other people during exams and quizzes. You may discuss the homework assignments with other students, but you should make sure that you understand the responses. If you simply type in the answer that someone else tells you, you lose a valuable opportunity to test your understanding.

Final Exam Policy. If extenuating circumstances occur, students may submit a Final Exam Reschedule request for review by the Associate Provost. Students may not negotiate a make-up date directly with the course instructor. Students may request to reschedule an exam only under the following circumstances:

- Two final exams scheduled at the same time;
- Three final exams scheduled in one calendar day; one of the exams will be rescheduled at the convenience of the instructor and the student;
- Serious illness, or personal emergency; the student is required to present documentation to validate.

The final exam schedule is visible on the Registrar's website by the beginning of each semester (<https://drew.edu/academic/office-of-the-registrar/registration/final-exam-schedule/>). Students are expected to schedule travel plans to occur AFTER their final exams.

Grade Cutoffs: We will use the following scale in determining your course grade from your course average. The course grades are not curved. Final grades are assigned according to the standard scale: A = 92 to 100; A- = 90 to 92; B+ = 88 to 90; B = 82 to 88; B- = 80 to 82;

C+ = 78 to 80; C = 72 to 78; C- = 70 to 72; F = 0 to 70 (if registered as an undergraduate: class D+ = 68 to 70; D = 62 to 68; D- = 60 to 62; F = 0 to 60).

Academic Accommodations. Your experience in this class is important to me. If you have already established accommodations with the Office of Accessibility Resources (OAR), please provide me with a copy of your accommodation letter at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through the Office of Accessibility Resources (OAR), but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are encouraged to contact OAR. OAR offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions.

Although a disclosure may take place at any time during the semester, students are encouraged to do so early in the semester, because, in general, accommodations are not implemented retroactively. Office of Accessibility Resources contact information:

Director-Dana Giroux

Location-Brothers College, Room 119B

Phone: 973-408-3962

Email: dgiroux@drew.edu, disabilityserv@drew.edu

Academic Integrity Policy. All students are required to uphold the highest academic standards. Any case of academic dishonesty will be dealt with according to the guidelines and procedures outlined in Drew University's "Standards of Academic Integrity: Guidelines and Procedures," which is located in the academic policies section of the catalog (<https://drew-caspersen-catalog.coursedog.com/academicpolicies#academic-integrity-policy1>).

Generative AI. I encourage you to explore Artificial Intelligence (AI) resources for learning purposes. When using AI, you must (1) fully document and cite all material (including any code, graph, formula, table, mathematical equation, etc.) that was not generated by you, (2) check information generated by AI and take full responsibility for its accuracy, and (3) be able to identify where and how you used any AI tools and how they contributed to your work. You are not allowed to use AI for any quizzes or exams. Use of any AI tools without permission is unacceptable and will be reported as an academic integrity violation. If you have any doubts about what is acceptable please discuss them with the professor.

Supporting Student Success, Center for Academic Excellence. All Drew students can access subject tutoring, writing support and academic coaching free of charge in the Center for Academic Excellence (CAE), located in the library. Seeking help through learning support resources in the CAE can help you achieve your academic goals. To access the appointment schedule, please visit drew.mywconline.com and follow the instructions on the landing page; if a tutor is not available, please submit the Tutor Request form (https://docs.google.com/a/drew.edu/forms/d/e/1FAIpQLScnfIAm1a9rdfa2s_fjrdW6tVdYtn04biG72acpWT9HixYt1w/viewform). For any other questions, email cae@drew.edu.

Preliminary Schedule (subject to change)

Date	Contents	Notes
8/26	Introduction & Getting started with R	
9/2	Ch2: Univariate distribution	No class on Mon; Quiz 1 (Wed)
Last day to drop without “W” is Monday, 9/8		
9/9	Ch3: Location, spread, and skewness	Quiz 2 (Wed)
9/16	Ch4: Transformation	Quiz 3 (Wed)
9/23	Ch4: Z-scores	Exam 1 (Mon)
9/30	Ch5: Bivariate relationships	Quiz 4 (Wed)
10/7	NO CLASS (Fall Break)	
10/14	Ch5: Bivariate relationships (cont'd)	Quiz 5 (Wed)
10/21	Ch6: Simple linear regression	Quiz 6 (Wed)
10/28	Ch7: Basic probability; Ch8: Normal distr.	Exam 2 (Mon)
11/4	Ch9: Central Limit Theorem	Quiz 7 (Wed)
11/11	Ch11: One sample inference	Quiz 8 (Wed)
Last day to drop with “W” is Tuesday, 11/18		
11/18	Ch11: Independent and paired samples	Quiz 9 (Wed)
11/25		Exam 3 (Mon)
12/2	Ch13: ANOVA	ANOVA quiz (Wed)

Based on the university schedule, our final exam is scheduled for **Tuesday, Dec 9, from 7:30 to 10:30 pm.**